Pattern Recognition

Neural Style Transfer

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1 Problem Statement

In this project, our objective is to implement a neural style transfer pipeline that takes two input images and a style threshold:

- A content image, such as a photograph of a city or a person
- A style image, such as a painting by Van Gogh or Picasso
- The style threshold is a scalar value that determines the degree to which the style is transferred onto the content image

Your model will generate a new image that preserves the content of the first image while adopting the style of the second, modulated by the threshold value. To understand the underlying principles of neural style transfer, we follow the approach of the foundational paper on this topic.

2 Data Loading and Splitting

From the provided datasets, 7000 content images were sampled from the Google-scraped image dataset and 3000 style images were sampled from the WikiArt dataset.

Afterwards, a stratified split was applied to the image paths to relocate the sampled images to another directory.
The following are the number of images in each of the training, validation, and test sets:

	Content Images	Style Images
Training Set	5600	2366
Validation Set	700	295
Test Set	700	322

The image paths were loaded into a pytorch Dataset, then each Dataset was loaded into a Dataloader for batch processing and lazy loading to avoid memory issues.

3 Data Processing

For this step, a transform was composed to be applied when loading images. Each image would be resized to 256x256 and the pixel values would all be normalized for faster model training.

4 Creating the Model

A pre-trained VGG19 model was utilized as an encoder for the network and a decoder was created from scratch using pytorch.

5 Fine-tuning the Model

The model was first trained using content and style images from the training set. Then the optimal threshold was chosen as a hyperparameter and evaluated on the validation set. Finally, the model was evaluated on the test set.

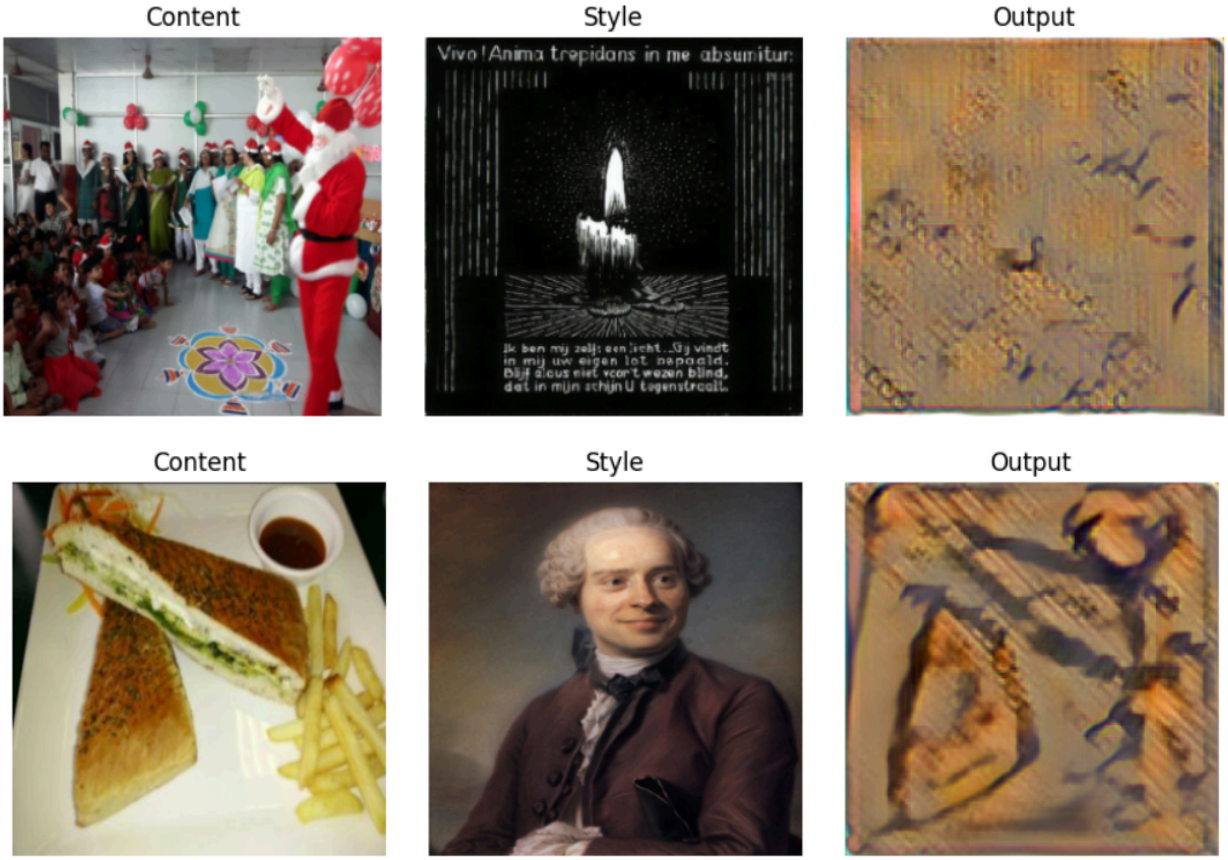
The optimal styling threshold was found to be 0.9 when evaluated on the validation set.

5.1 Test Performance

When evaluated on the test set with the optimal threshold, the model has the following performance:

- Content Loss: 26.210662
- Style Loss: 7.818716e+06
- Total Loss: 7.818742e+06

5.1 Test Image Outputs



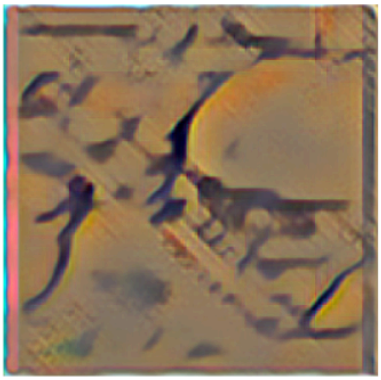
Content



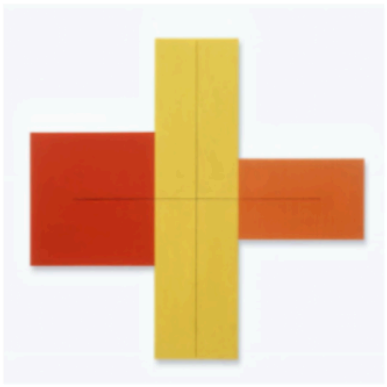
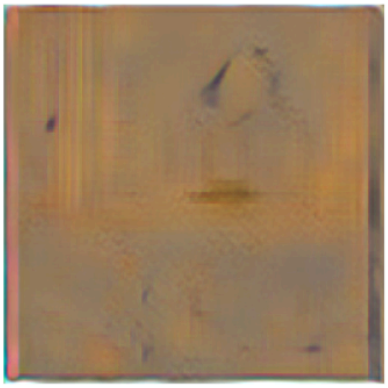
Style



Output

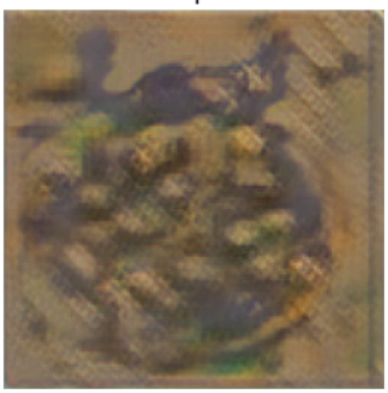


Content | Style | Output

Content | Style | Output



Content | Style | Output





Content



Style



Output



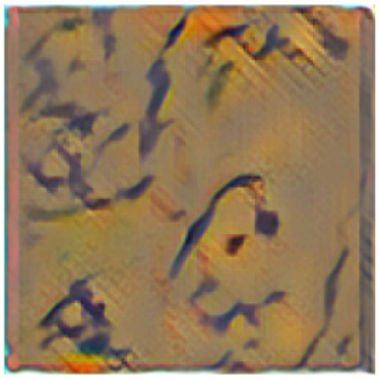
Content



Style



Output



Content



Style



Output

